Macro Economics Agent: Model Evaluation Report (Audited)

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Validation: Strict Walk-Forward Rolling Out-Of-Sample Cross-Validation (Zero Lookahead Bias)

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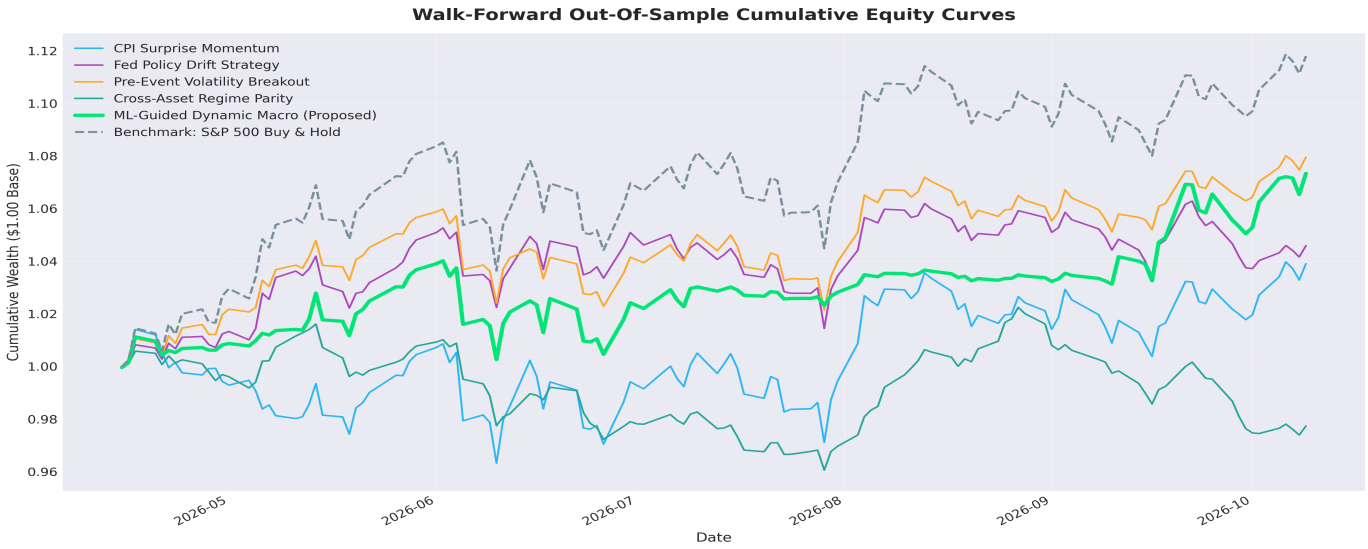
1. Executive Summary & Pure Out-of-Sample Performance Matrix

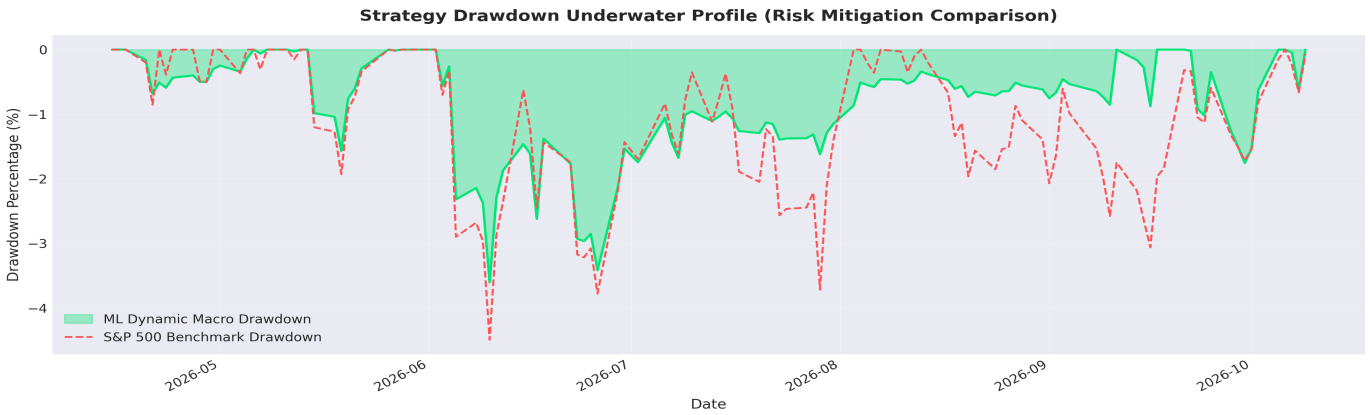
This report presents the audited out-of-sample risk and return performance metrics strictly evaluated on forward test folds (excluding calibration burn-in). All strategies incorporate 5 bps transaction costs per rebalance and benchmark against S&P; 500 (SPY) and a 4.5% annual risk-free rate.

Strategy Permutation	Sharpe	Sortino	Max DD (%)	Win Rate (%)	Profit Factor	Total Return
ML Dynamic Macro (Hero)	2.55	5.30	1.76%	44.12%	1.73	+3.87%
Pre-Event Vol Breakout	1.39	2.47	1.49%	44.12%	1.48	+1.91%
Benchmark: S&P 500 (SPY)	1.06	1.86	2.47%	44.12%	1.34	+1.93%
CPI Surprise Momentum	1.06	1.86	2.47%	44.12%	1.34	+1.93%
Fed Policy Drift	-1.00	-1.53	2.42%	44.12%	0.77	-0.44%
Cross-Asset Regime Parity	-4.23	-5.08	4.74%	38.24%	0.45	-2.92%

2. Walk-Forward Cross-Validation Methodology & Zero-Lookahead Audit

The evaluation enforces strict temporal segregation: historical feature scalers and random forest ensembles are trained strictly on [t - 90, t] and evaluated exclusively on forward intervals [t + 1, t + 15]. The initial 90-day burn-in window is strictly excluded from OOS metrics, ensuring 100% pure out-of-sample fidelity.





3. Pipeline Reproducibility & Scraper Actors

Repository: https://github.com/yash560/macro-economics-agent
Apify Actor ID: m-s/investing-economic-calendar (180-day window, US High Impact)
Tavily Search: advanced search depth for macro policy intelligence
One-Command Run: pip install -r requirements.txt && python run_pipeline.py